

Synthetische Daten und Testdatengenerierung

*Sammelst du noch
oder generierst Du schon?*



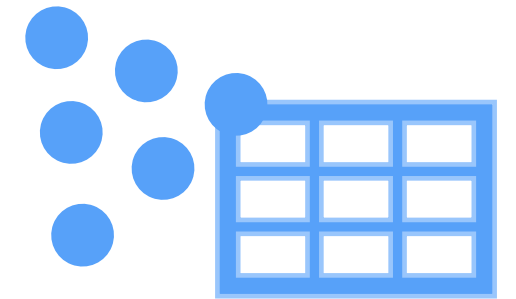
Gerhard Svolba
Data Scientist
SAS Austria



What is synthetic data?

Synthetic data is generated by applying a sampling technique to real-world data or by creating simulation scenarios where models and processes interact to create completely new data not directly taken from the real world.

– *Gartner*



Synthetic data and generative AI

Synthetic data has become important with the emergence of GenAI given GenAI models require large amounts of data for training.

Synthetic data is also considered to be a privacy preservation enabler.

Synthetic data generation can be considered a subset of GenAI given it generates data.



Why do we need synthetic data?

We have
no access



No Internet Connection

We are not allowed to
use the data

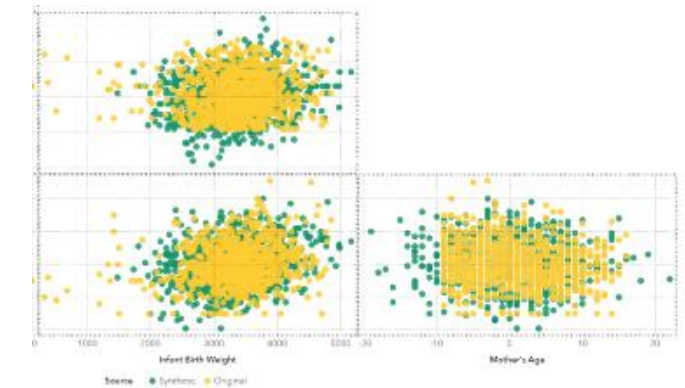


Data do not
exist (yet)

404
Not Found

The resource requested could not be found on this server!

Data is not available in
sufficient quantity



Consequence: extended lead times, high effort
and projects cannot be initiated



Synthetic data helped improve AI model accuracy, potentially reducing losses.

PROBLEM

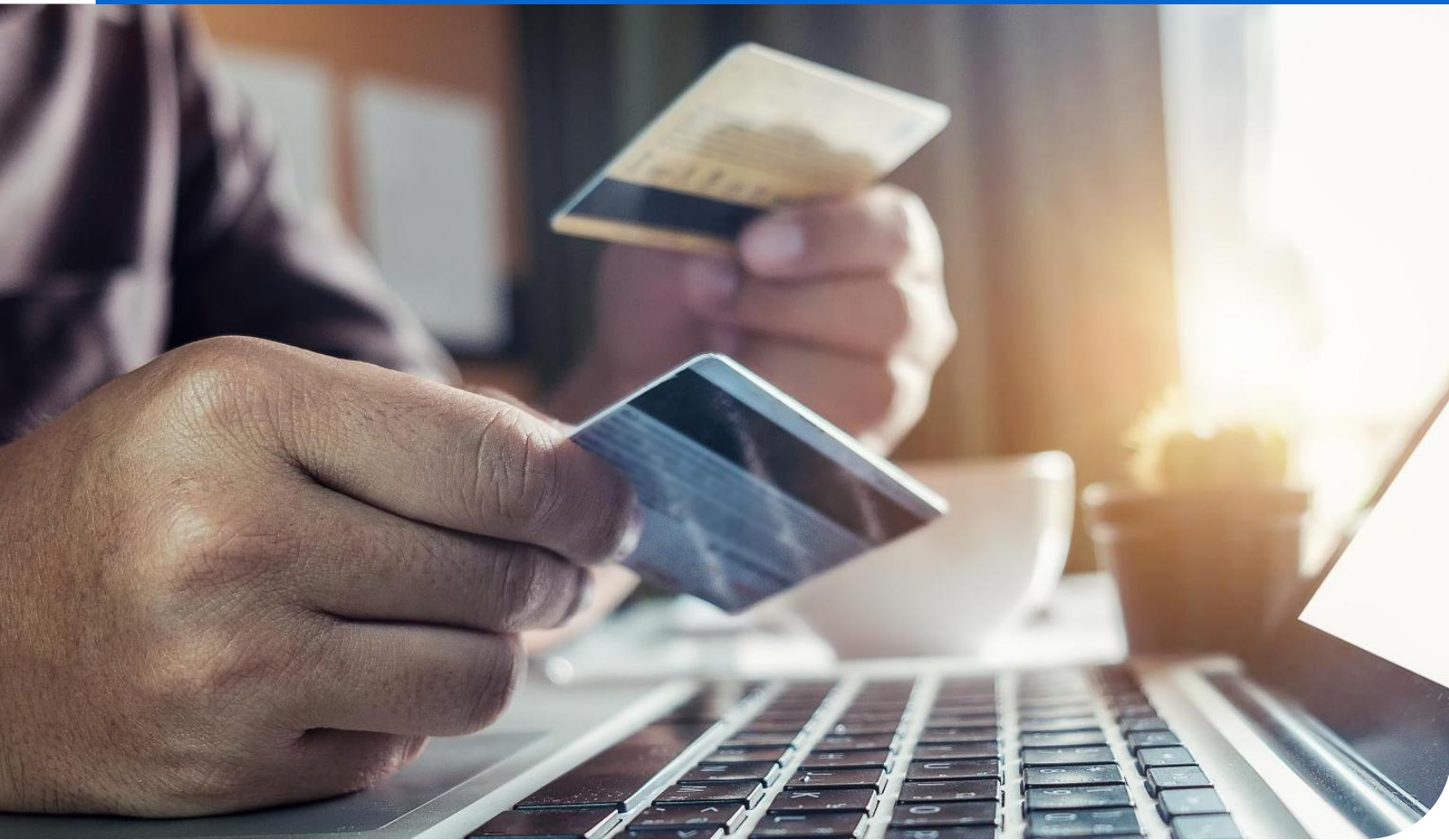
- Machine learning credit scoring models must guide personal loan decisions in a **fair, responsible manner**
- Previous models using only real-life data faced **limitations in accuracy, scalability, and data sensitivity**

ACTION

- **Tested synthetic data generated with SAS** to improve machine learning model, comparing model accuracy versus models using only real-life data

RESULT

- **28% improvement** in model accuracy with facilitated machine learning efforts, supporting potential reduction in losses



Synthetic data enhanced customer privacy and collaboration with external vendors

PROBLEM

- Lack of safe and realistic test data. Privacy leakage concerns.
- Needed a way to share data with external vendors without risks to customer privacy and compliance.

ACTION

- Generated synthetic data repository for on-demand test data.
- New products were generated in a sandbox with synthetic data.

RESULT

- Sandbox was safely accessed by over 700 external developers.
- Enhanced privacy, reduced risk, improved collaboration with external vendors.

Organization: US Health Care Provider
Industry: Health Care
Region: United States



Synthetic data unlocks value from sensitive data without compromising privacy and security

PROBLEM

- Patient healthcare data is protected by privacy regulations, restricting access to data and increasing costs.
- These costs made performing analytics with the data cost-prohibitive.
- Provider must support an ecosystem of shared research and innovation involving universities and research groups.

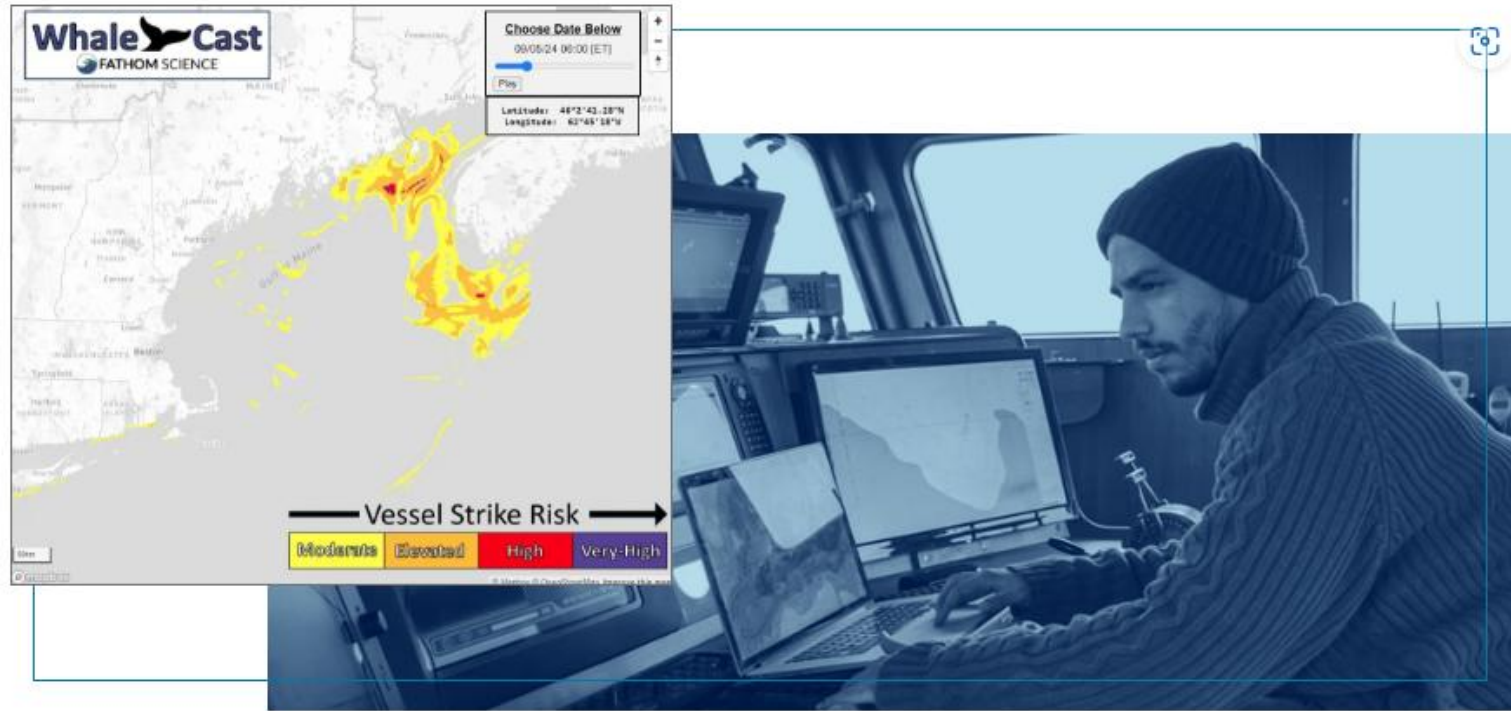
ACTION

- Provide secure and unified platform which enables synthetic data generation to simulate patient behavior and outcomes, test treatment plans and choose optimal care paths.
- Platform also offers critical supporting capabilities such as data transformation, generation and post-processing.

RESULT

- Enables patients to benefit from data-driven research on synthetic clinical datasets
- Reduced risk to customer privacy

Organization: Fathom Science
Industry: Marine Conservation
Region: United States



Synthetic data
augmented insufficient
real data, helping
conserve right whales

PROBLEM

- North Atlantic right whales are endangered, partly due to being struck accidentally by boats
- Need a predictive heat map of whale movement to inform ship captains
- Insufficient whale sighting data to validate predictive model of whale movement

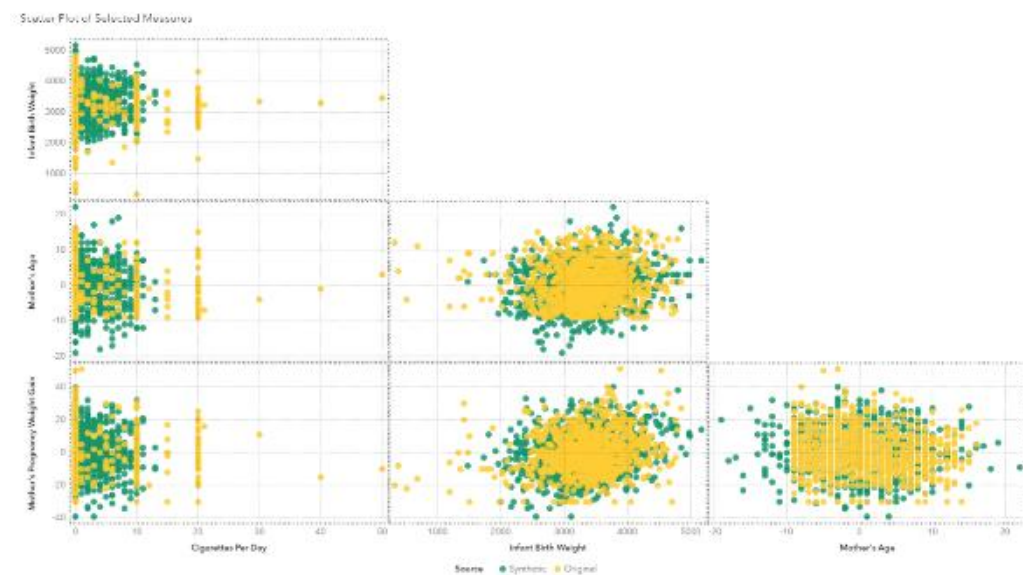
ACTION

- Used SAS to generate synthetic data to achieve sufficient data for validation (expanded set from 40K to 500K data points).
- Fathom was able to [validate their predictive model](#), increasing confidence using machine learning.

RESULT

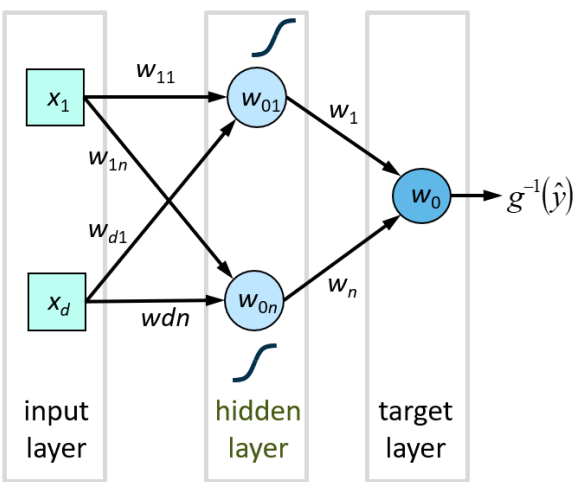
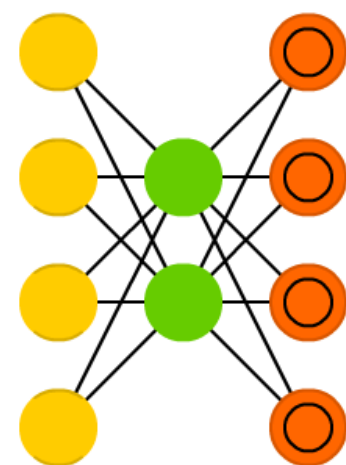
- WhaleCast tool can be integrated into existing boat on-board touch screens, helping mariners better understand areas with greater risk of striking whales

Frequently used methods for data synthetization



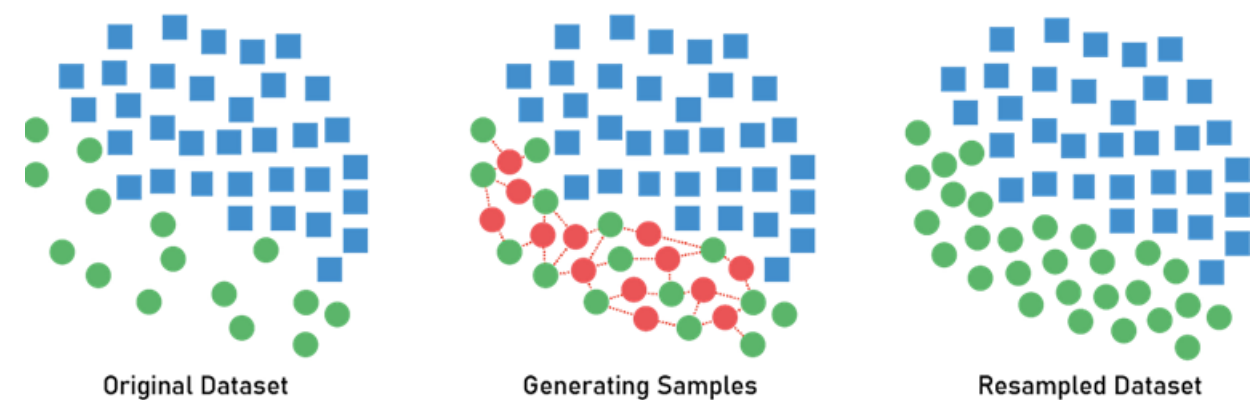
Multivariate Distributions

Auto Encoders



Generative Adversarial Networks

SMOTE - Synthetic Minority Oversampling Technique



kNN k-nearest-neighbour

SAS Viya Software Demo:

**Basic Example: Synthetic Data
with PROC SIMNORMAL**



Options View  Open  Save All SAS Studio compute co 13 SynthData Cars.sas x

+

 Run  Cancel    Copy to My Snippets  Code to Flow    Debug  Clear Log Jun 2...

Code

```
1  /*****>
2  *** SASHELP.CARS Data
3  ****>
4
5
6  title sashelp.cars Data;
7  ⊖ proc print data=sashelp.cars(obs=10);
8    var invoice horsepower mpg_city mpg_highway weight;
9  run;
10
11
12  ⊖ proc sgplot data=sashelp.cars;
13    scatter x=horsepower y=mpg_highway;
14  run;
15
16  title;
17
18
19
20
21  /*****>
22  *** Create Data Generator
23
```

Log Results

 Errors (0)  Warnings (0)  Notes (11)

There are no messages.

```
⊕ /* region: Generated preamble */
9
0
1  proc sgplot data=work.synth_cars;
2    scatter x=horsepower y=mpg_highway;
3  run;
OTE: PROCEDURE SGPLOT used (Total proces
  real time          0.10 seconds
  cpu time            0.07 seconds

OTE: Listing image output written to
  /opt/sas/viya/config/var/tmp/compsr
  c20a-2467-4f52-b2ef-d5309e54f797-23
OTE: There were 500 observations read fr
4
⊕ 5  /* region: Generated postamble */
6
```

Original and Synthetic Data

Original Cars Data

Obs	Invoice	Horsepower	MPG_City	MPG_Highway	Weight
1	\$33,337	265	17	23	4451
2	\$21,761	200	24	31	2778
3	\$24,647	200	22	29	3230
4	\$30,299	270	20	28	3575
5	\$39,014	225	18	24	3880
6	\$41,100	225	18	24	3893
7	\$79,978	290	17	24	3153
8	\$23,508	170	22	31	3252
9	\$32,506	170	23	30	3638
10	\$28,846	220	20	28	3462

Synthetic Cars Data with PROC SIMNORMAL

Obs	Invoice	Horsepower	MPG_City	MPG_Highway	Weight
1	\$43,270	262	22	29	3439
2	\$15,541	127	26	32	4039
3	\$35,198	264	13	21	3741
4	\$27,475	145	28	35	2980
5	\$31,493	267	19	25	3426
6	\$16,615	210	24	28	3814
7	\$22,891	180	14	20	3670
8	\$12,328	138	23	30	3294
9	\$25,389	172	24	30	2985
10	\$16,868	220	18	24	3937

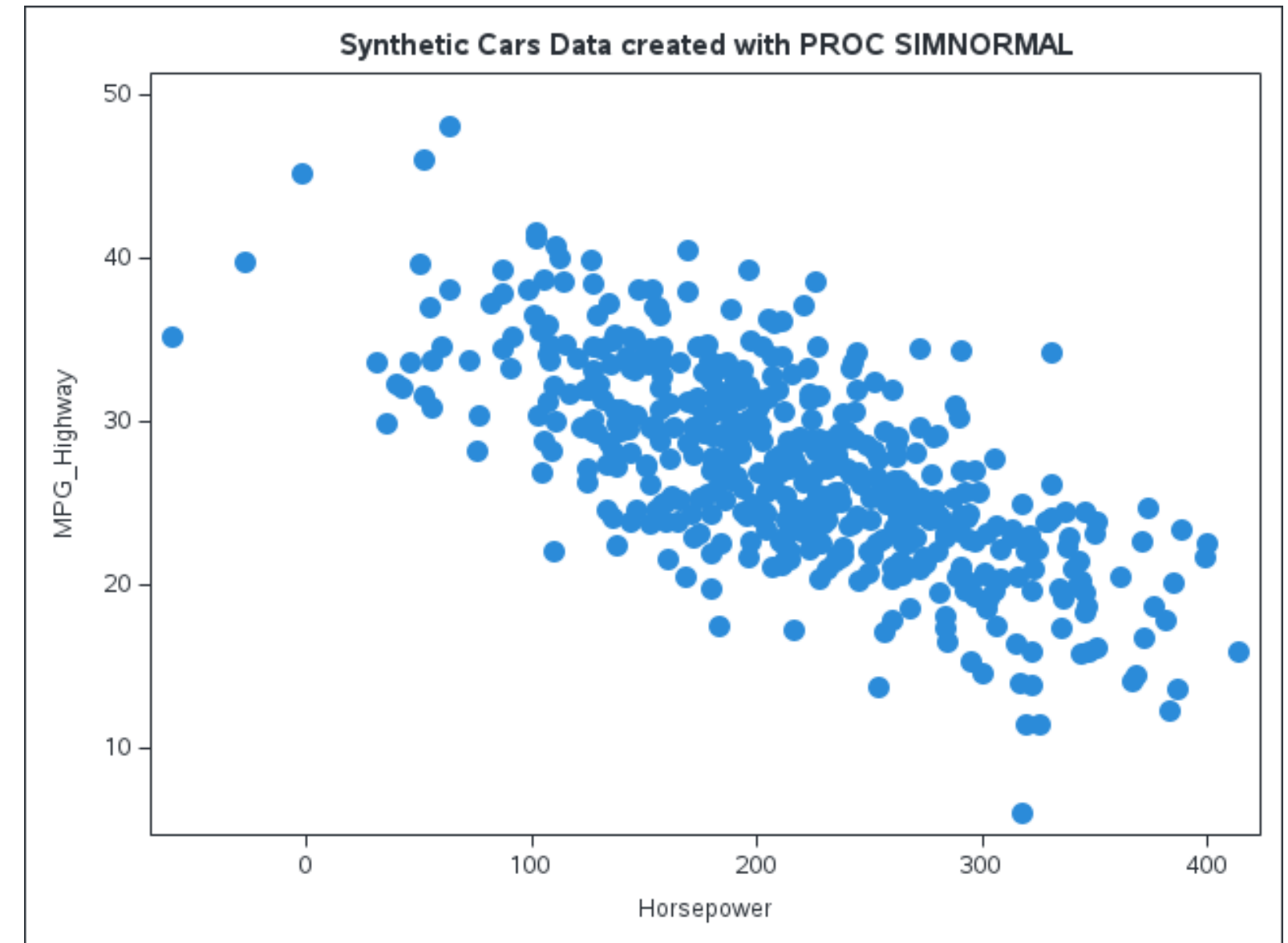
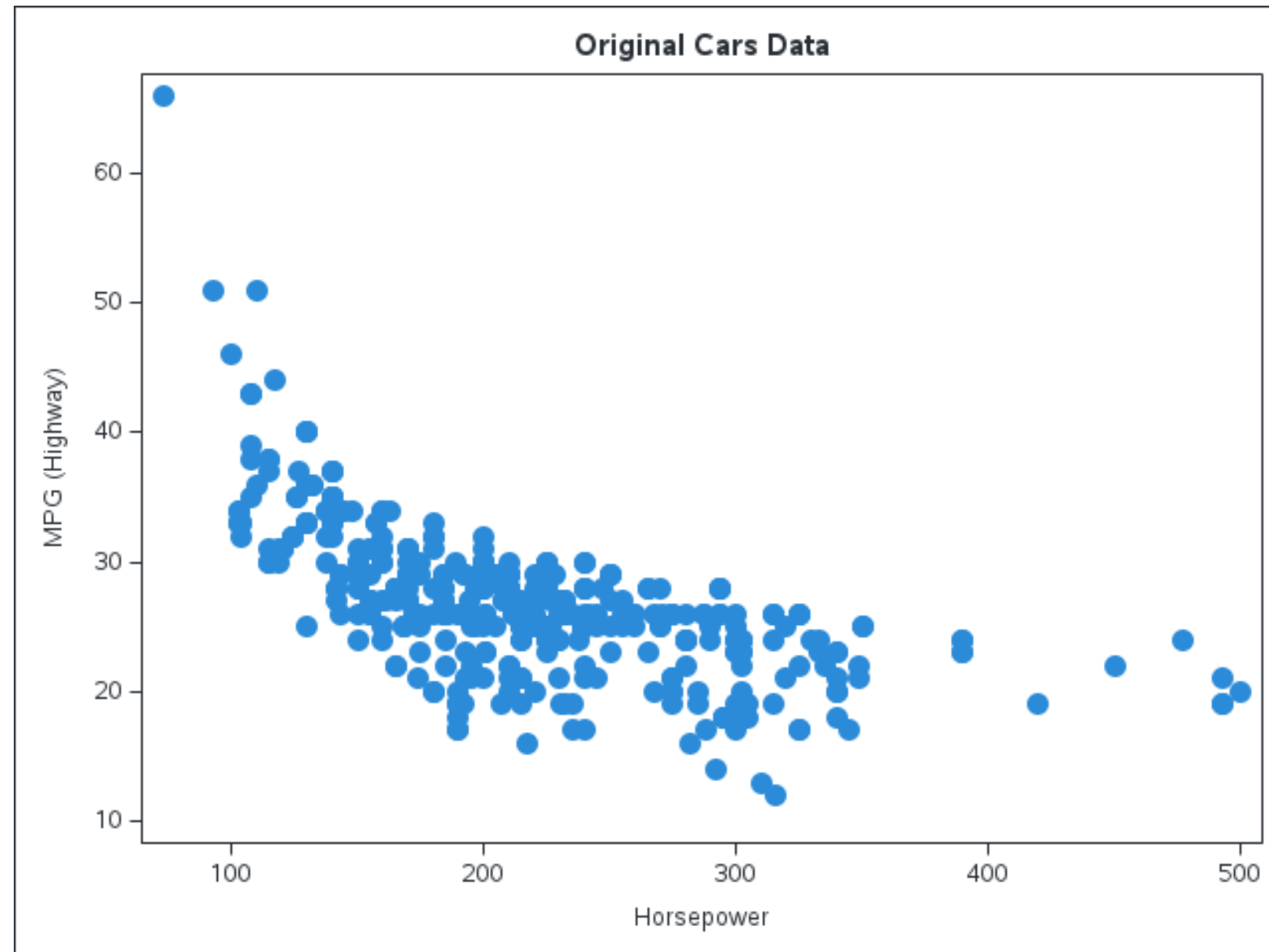
PROC CORR and PROC SIMNORMAL

```
proc corr data=sashelp.cars out=work.cars_cov cov noprint nocorr;
  var invoice horsepower mpg_city mpg_highway weight;
run;

/*****
***   Use Generator to Create Synthetic Data
*****/

proc simnormal data=work.cars_cov(type=cov)
  out = work.CarsSynth_Simnormal
  numreal= 500
  seed = 1333;
  var invoice horsepower mpg_city mpg_highway weight;
run;
```

Scatterplot for Horsepower and MPG



Points to consider

- Categorical variables
- Non-linear relationships
- More than one table (relational schema)
- Time series data
- Privacy preservation
- Validation statistics (created automatically)

SAS Viya Software Demo:

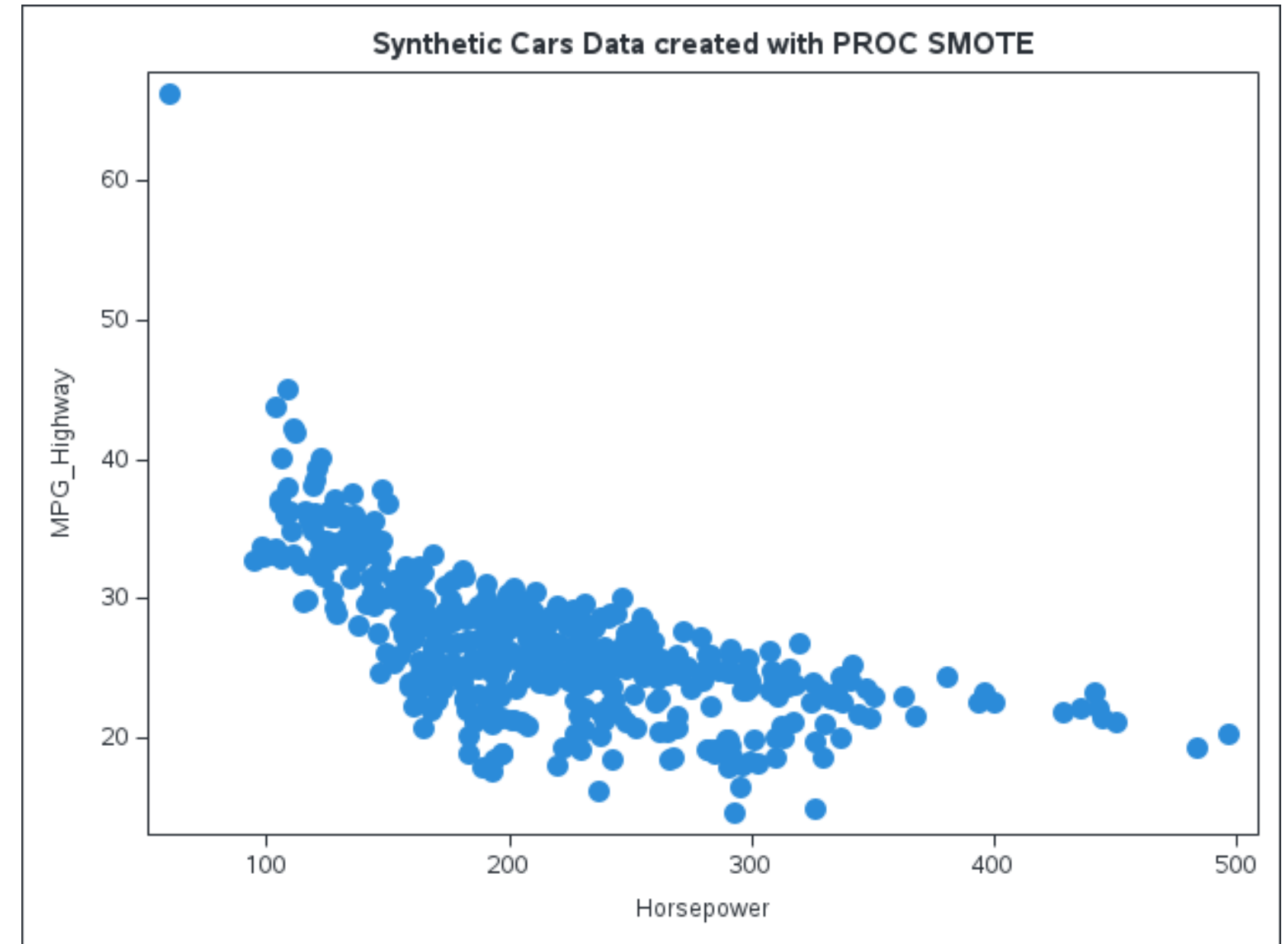
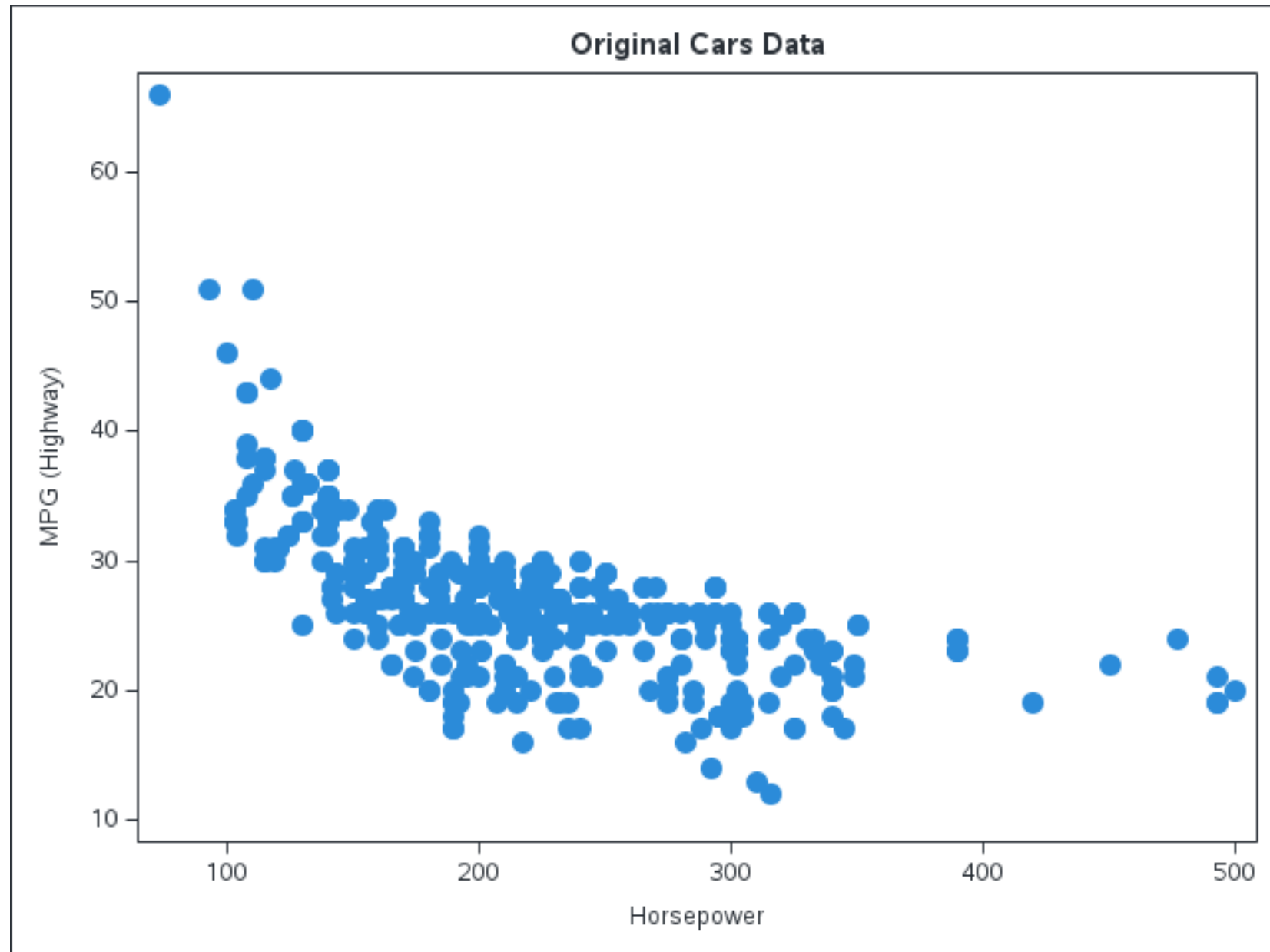
**Basic Example: Synthetic Data
with PROC SMOTE**



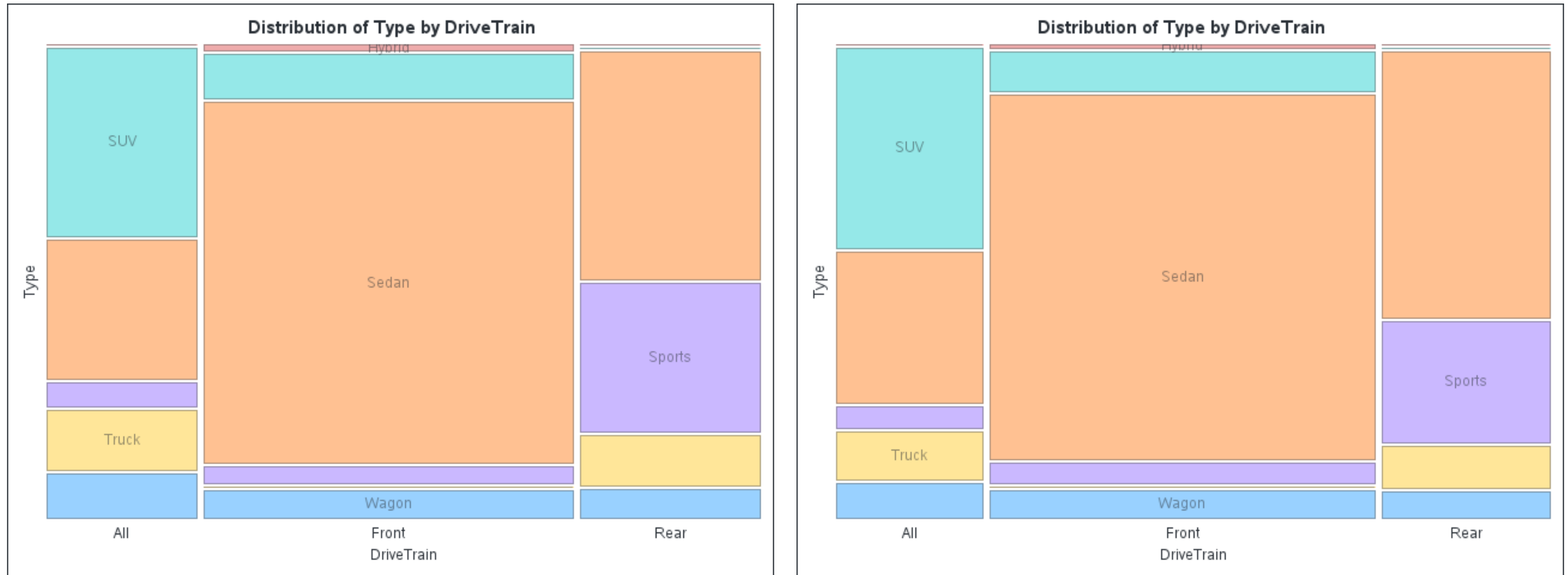
PROC SMOTE

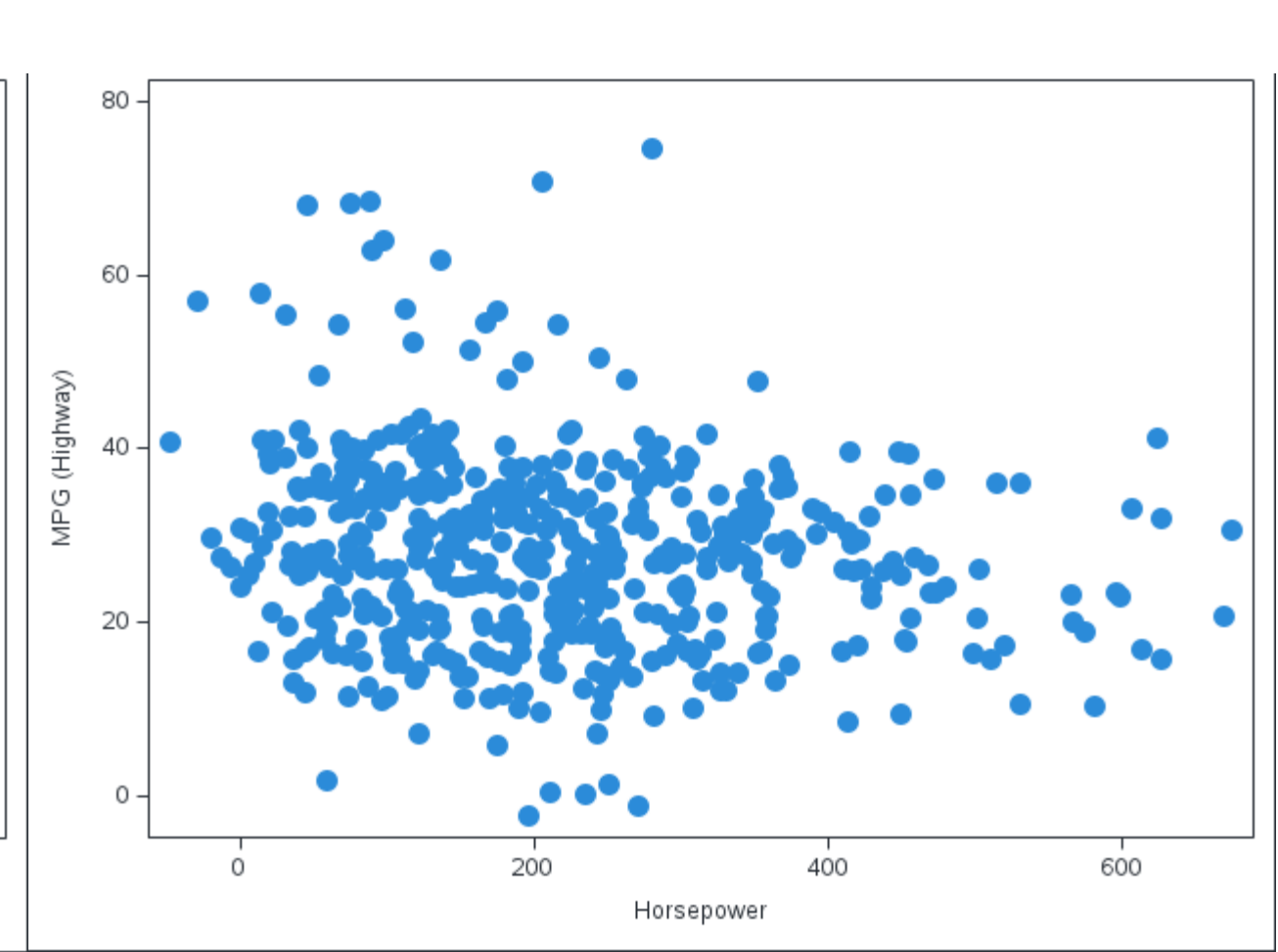
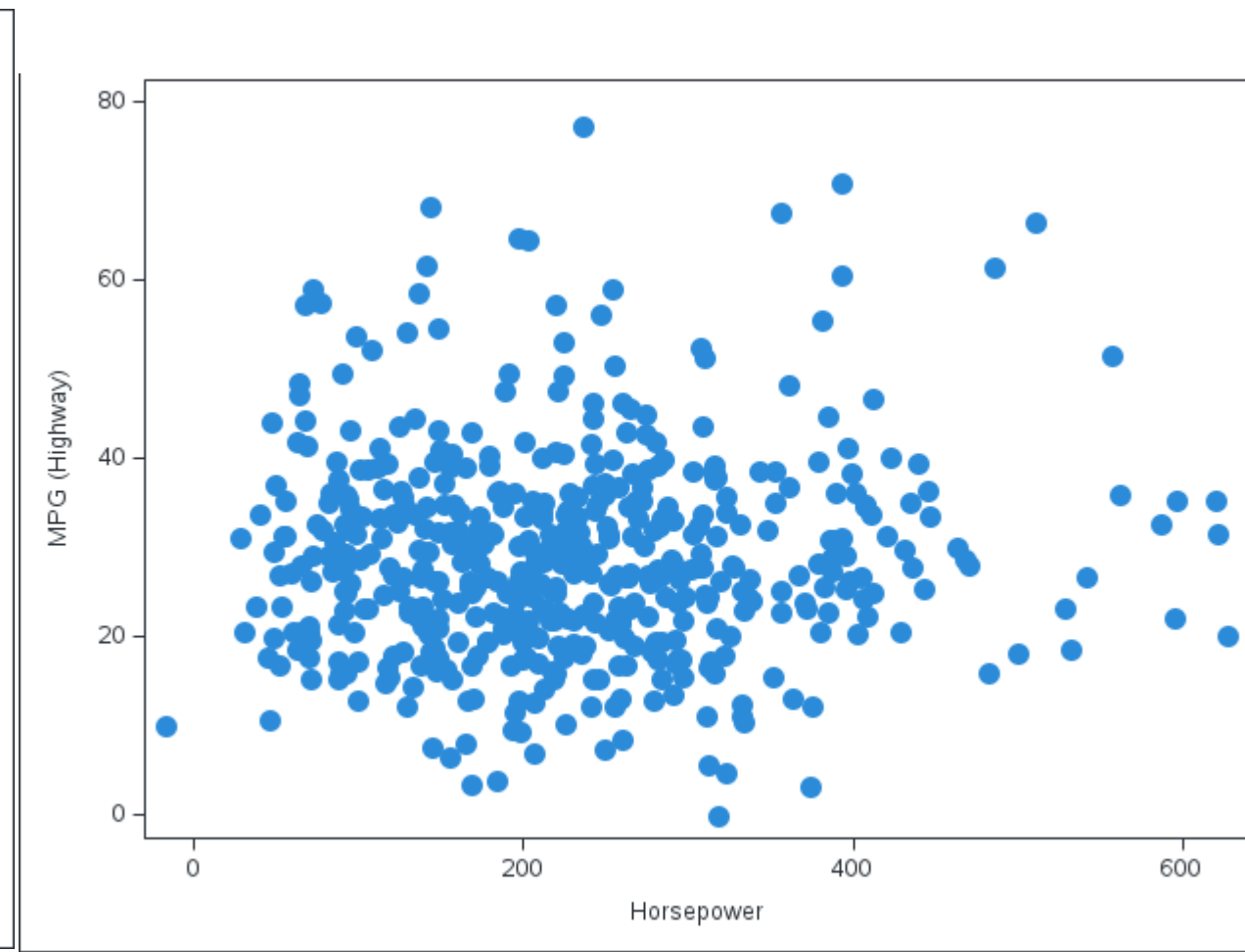
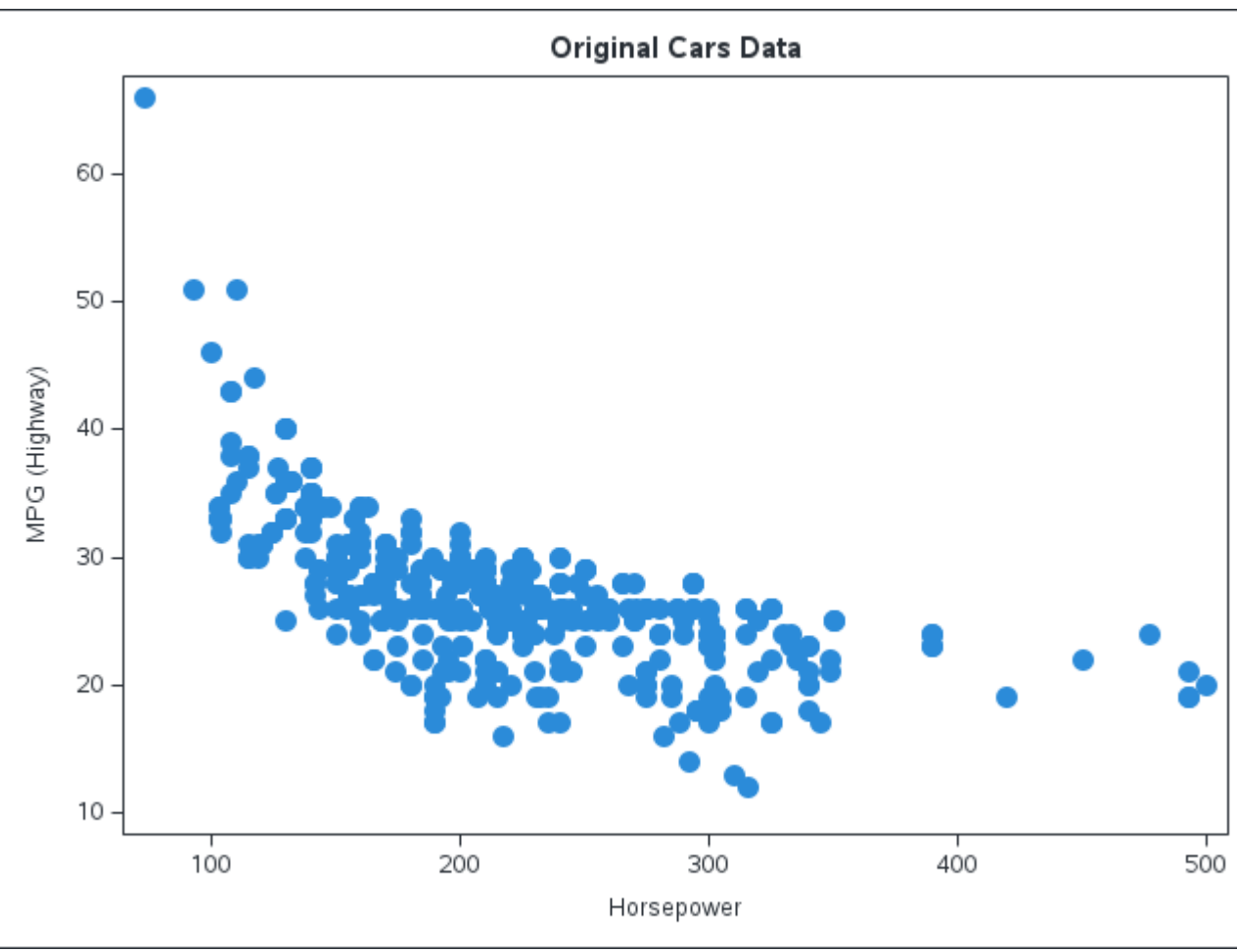
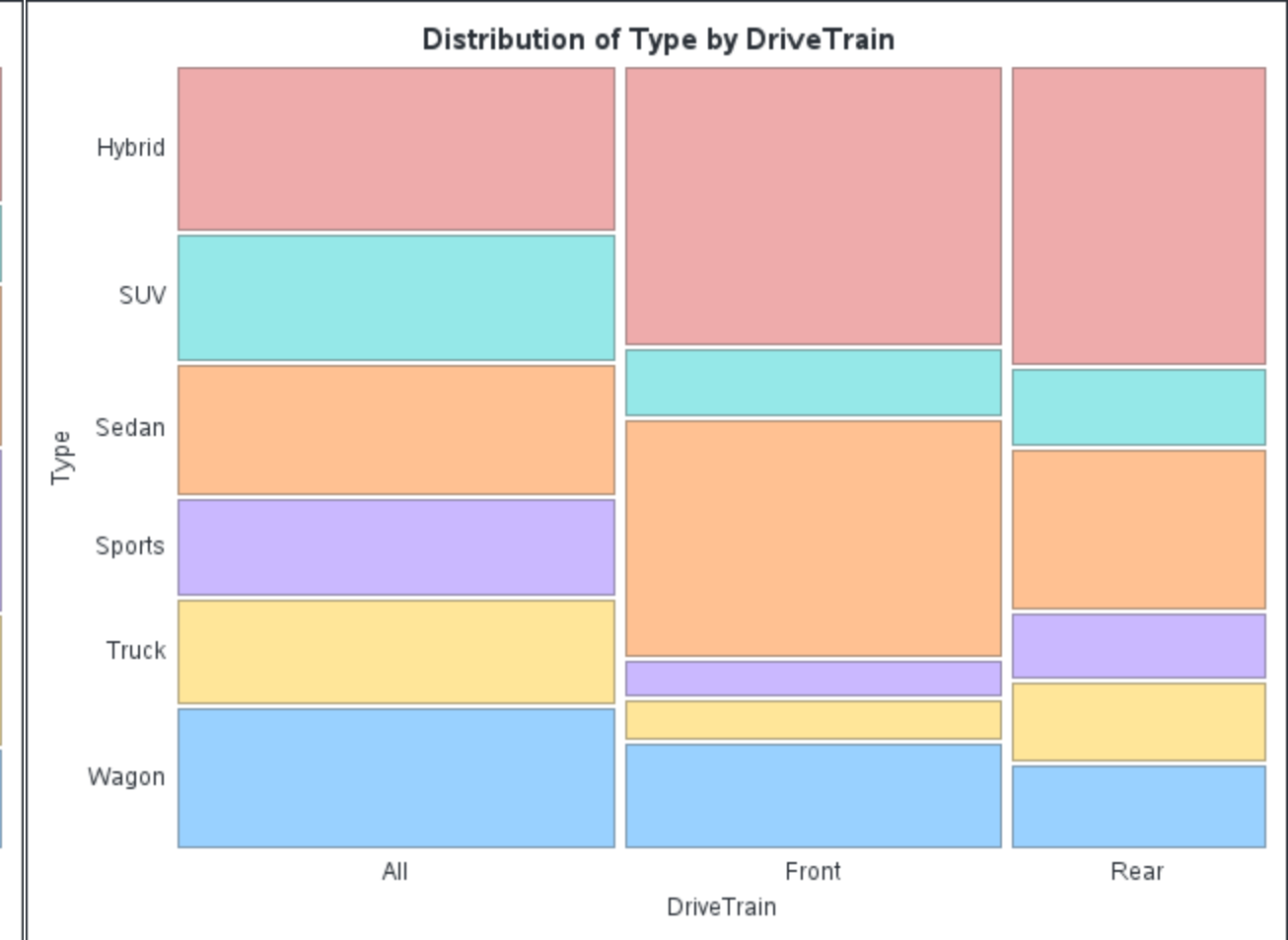
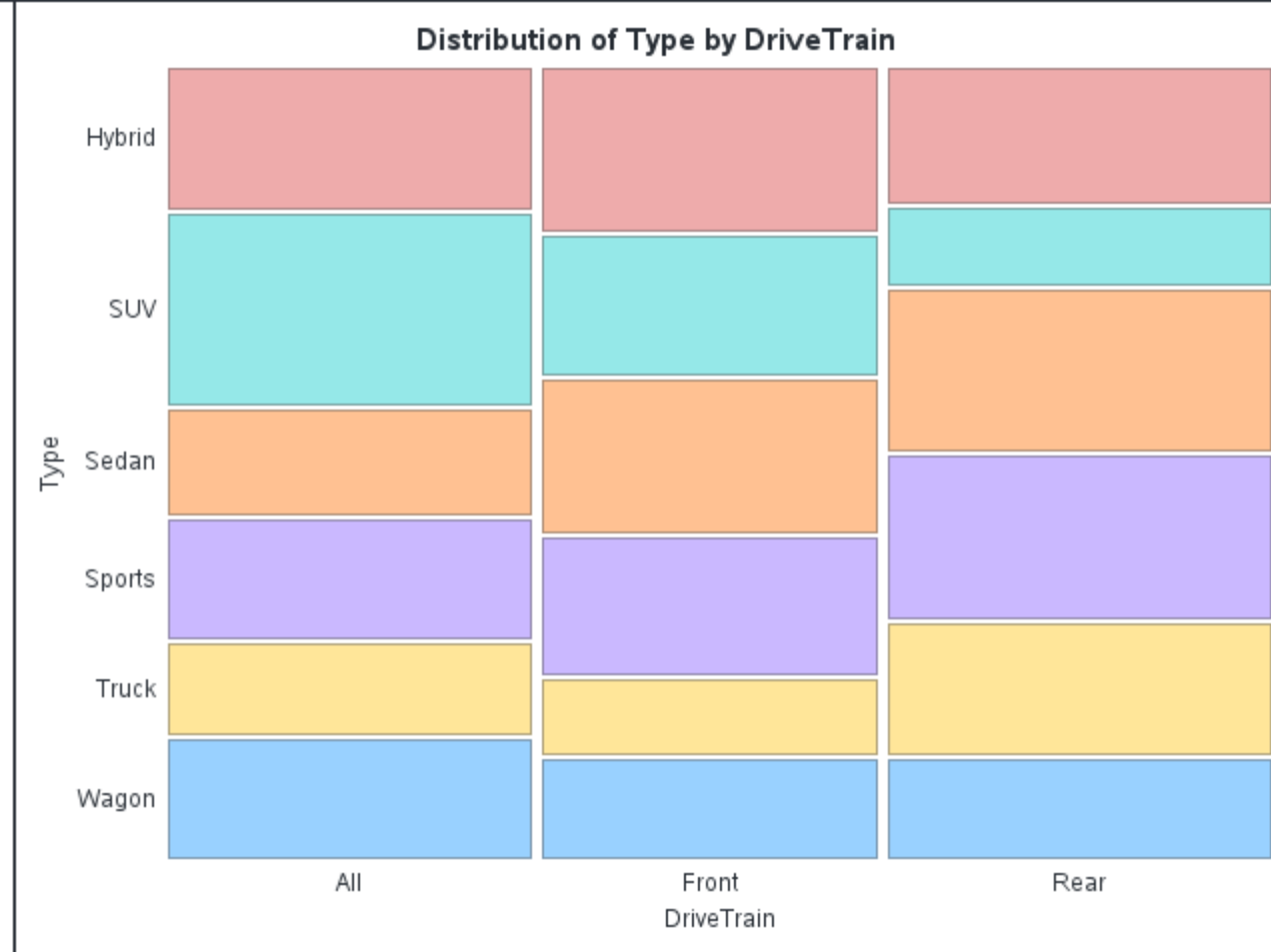
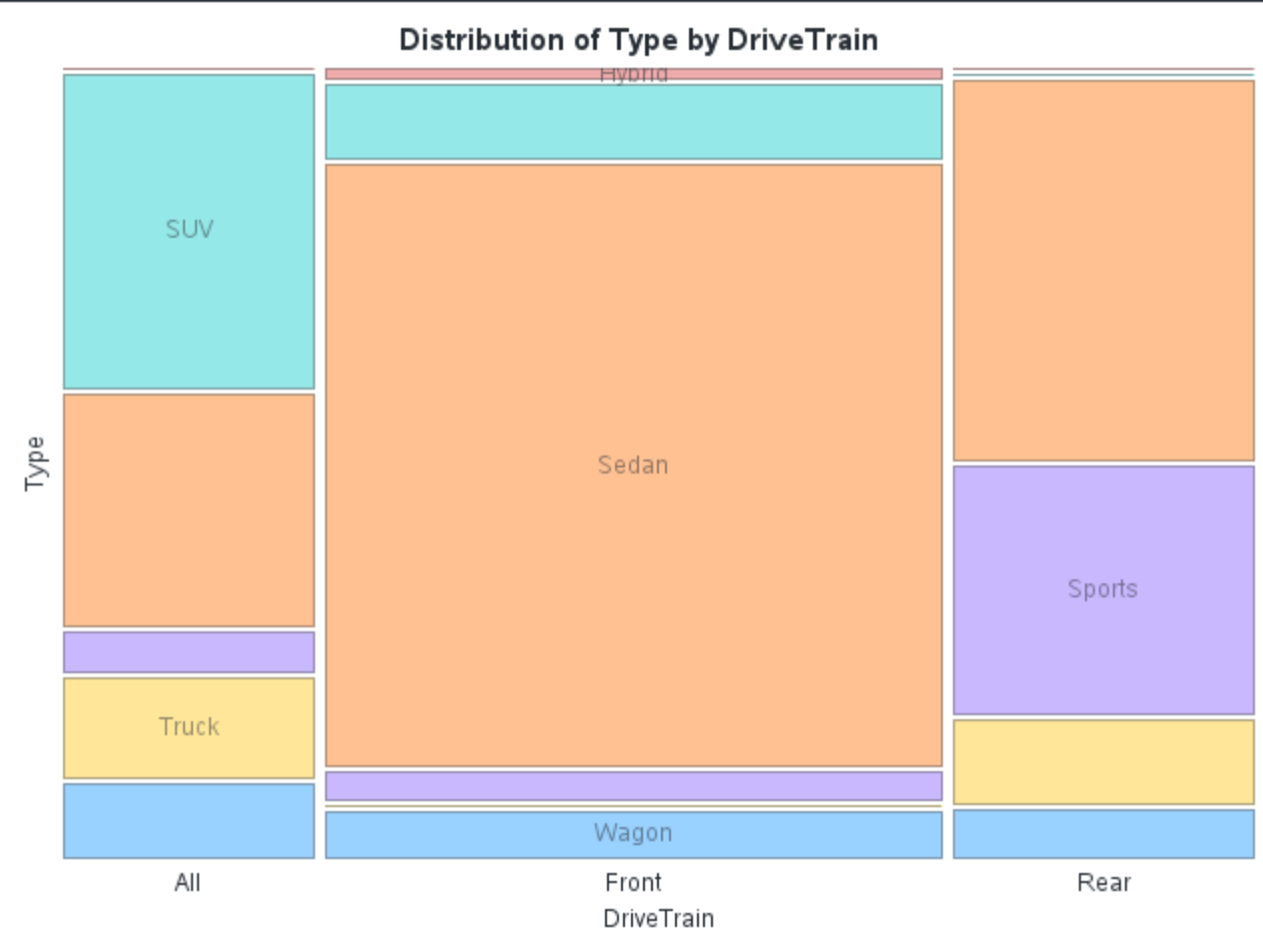
```
proc smote data = casuser.cars seed=1;  
  input type drivetrain/level = nominal;  
  input invoice horsepower mpg_city mpg_highway weight;  
  output out=casuser.CarsSynth_Smote;  
  sample numsamples=500  
    EXTRAPOLATIONFACTOR = 0.1  
    K = 7;  
run;
```

Scatterplot for Horsepower and MPG



Mosaic Plot for DRIVETRAIN and TYPE





SAS Viya Software Demo:

**Creation of Synthetic Data using
the SAS DATA MAKER**





SAS® Data Maker

Your gateway to seamless synthetic data generation

New project

Projects

Search

🔍



Create a project to get started.

New project

How to



Overview

See an overview of SAS Data Maker.

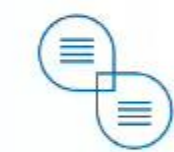
[Watch video](#)



Get Started

Learn about SAS Data Maker.

[View documentation](#)



Provide Feedback

We would love to hear about your experience.

[Submit feedback](#)



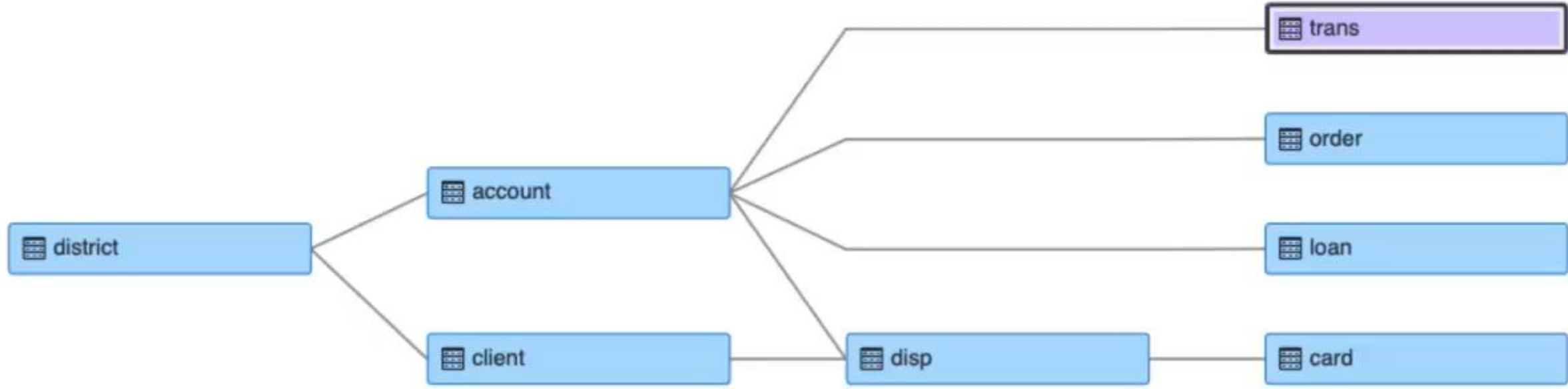
INNOVATE DEMO

- Source Data
- Training
- Evaluation
- Generate

Default model

- account
- card
- client
- disp
- district
- loan
- order
- trans

Structure Columns



trans

Source:
banking/trans.csv.gz

Table type
Sequential

Sequential id
account_id

Sort by

Primary key
trans_id

Foreign keys

Key: Target: account... ACCOUNT.acc...

+ Add foreign key

Date modified:
Dec 31, 1, 11:23 PM

Date created:
Apr 17, 2025, 5:20 PM

Size:
38.15 MB



≡ INNOVATE DEMO

Source Data Training Evaluation Generate

📄 All Tables

📄 account

📄 card

📄 client

📄 disp

📄 district

📄 loan

📄 order

📄 trans

Structure Columns

Table	Column	Semantic Type
account	account_id	ID
account	district_id	Foreign key
account	frequency	Category
account	date	Date time
card	card_id	ID
card	disp_id	Foreign key
card	type	Category
card	issued	Date time



≡

INNOVATE DEMO

Source Data

Training

Evaluation

Generate

🕒

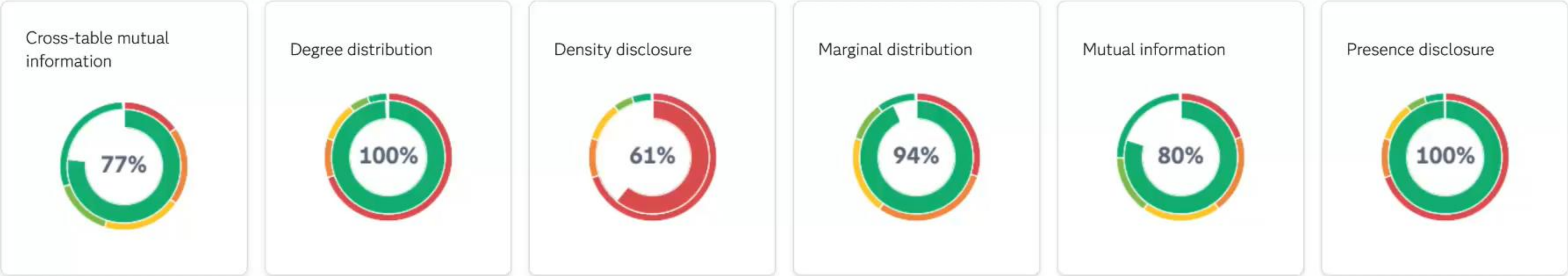
✓ 25 Apr 2025 at 14:55

▼

ⓘ

This model has been trained. Make a copy to train with different settings.

Make a Copy and Edit



Assessment Metrics

Generated Data Sample

Marginal distribution

Column

account.frequency (99.5%) ▼



Digital twin to generate synthetic sail races to train a decision support model for race tactics

```

/*****
*** PART II - START SIMULATION LOOP
*****/

*** Loop until epsilon around upwind mark has been reached;
do until ( DirectLine < &UpwindZone. or count > 200);

    count=count+1;
    racetime = racetime + &timestep.;

    if racetime <= &maxtime. then Winddir = input(RaceTime+WindSelectTime,wind_lkp.);
    else Winddir=180;

/*****
*** PART III - Analyse the Angle and decide about a FORCED tacking because of Layli
*****/

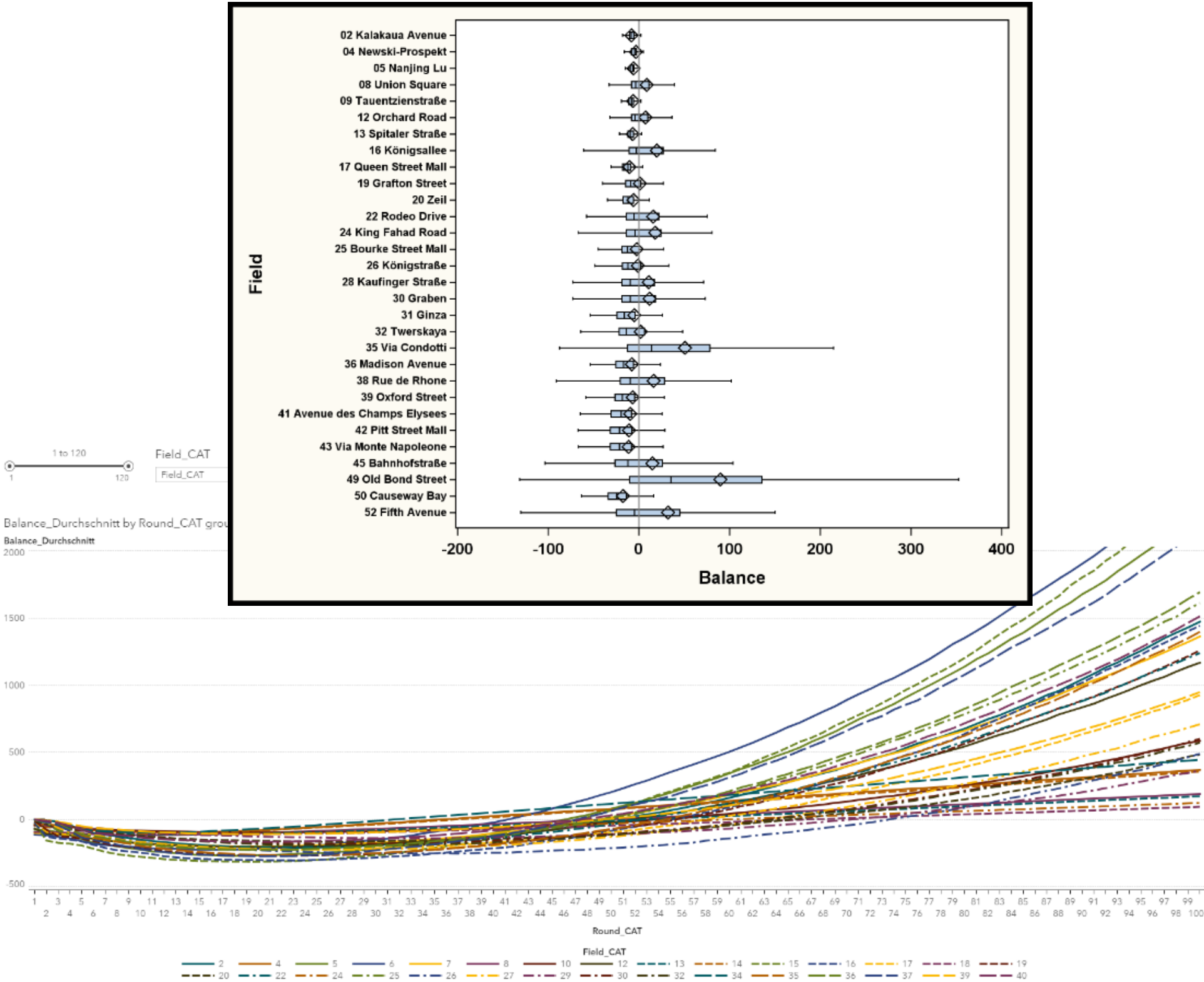
*** If AngleTack has already been set --> do nothing;
if AngleTack=1 and abs(AngleToMark) > &AngleTackLimit. then tack=0;
else do;
    *** If Layline exceeded --> TACK;
    if abs(AngleToMark) > &AngleTackLimit. then do;
        tack=1;
        AngleTack=1;
    end;
    *** otherwise continue;
else do;
    AngleTack=0;

/*****
*** PART IV - INDIVIDUAL BOAT TACTICS SCORING
*****/

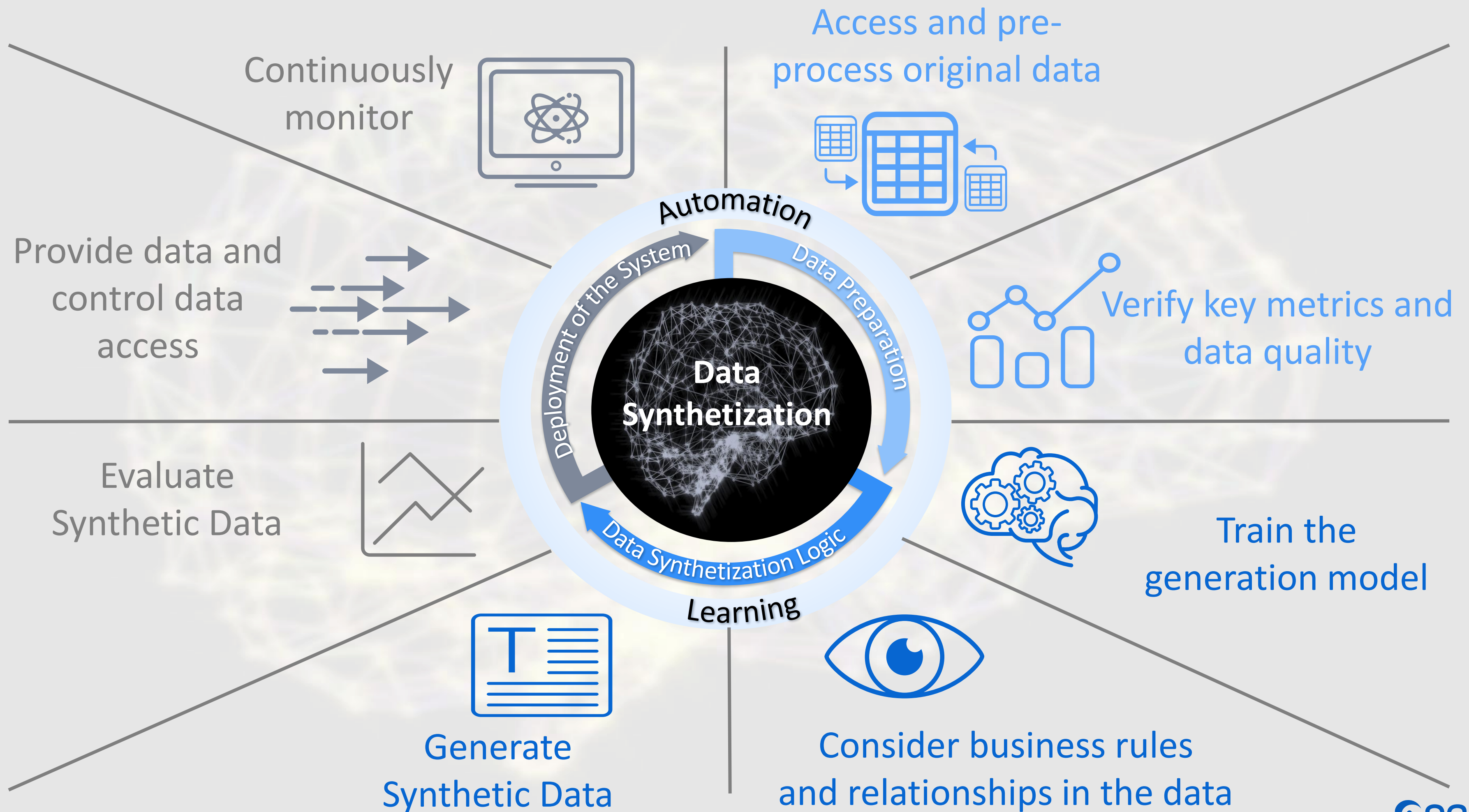
```



Digital Twin to Simulate Data for the Monopoly Board Game



You need more than just an analytical method!



Links

- Gerhard @ SAS Communities:
 - <https://communities.sas.com/t5/SAS-Communities-Library/Data-Science-and-Data-Preparation-Article-Overview-by-Gerhard/ta-p/727875>
- SAS Data Maker: https://www.sas.com/en_us/software/data-maker.html
- Proc SMOTE: https://documentation.sas.com/doc/en/pgmsascdc/default/casml/casml_smote_syntax04.htm#casml.smote.sample_K
- Whats new in SAS: <https://criptic.github.io/SAS-Whats-New/index.html>
- Monopoly Digital Twin
 - Analytics mal anders: E-Werk, Bahnhof oder Schlossallee? Der Monopoly-Check | SAS Forum Digital 2021
<https://www.youtube.com/watch?v=M1mlczR7KWI&list=PLdMxv2SumIKusTslCWoSJt06MoZyWazEU&index=3>